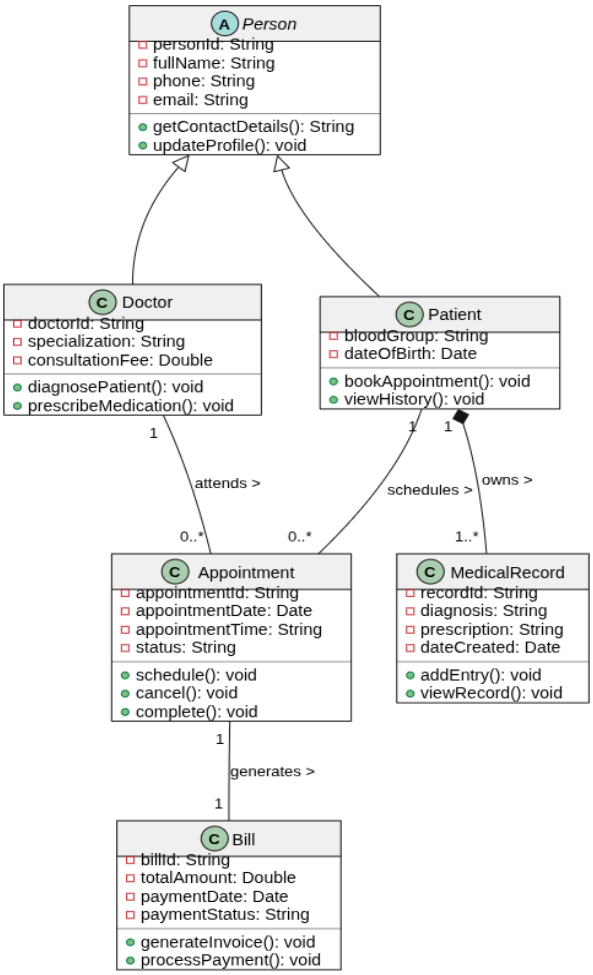
KIGALI INDEPENDENT UNIVERSITY (ULK)
DEPARTMENT OF COMPUTER SCIENCE / IT
COURSE: UML CLASS DIAGRAM MODELLING
INSTRUCTOR: ERIC MANIRAGUHA

REPORT 2: UML CLASS DIAGRAM REPORT

Project Title: Outpatient Clinic Management System (OCMS)
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Student ID: [202512802]
Date: October 2026

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1. COMPLETE UML CLASS DIAGRAM



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2. CLASSES, ATTRIBUTES, AND METHODS

1. Person (Abstract Class)
  - Attributes:
    * - personId: String
    * - fullName: String
    * - phone: String
    * - email: String
  - Methods:
    * + getContactDetails(): String
    * + updateProfile(): void

2. Patient (Subclass of Person)
  - Attributes:
    * - bloodGroup: String
    * - dateOfBirth: Date
  - Methods:
    * + bookAppointment(): void
    * + viewHistory(): void

3. Doctor (Subclass of Person)
  - Attributes:
    * - doctorId: String
    * - specialization: String
    * - consultationFee: Double
  - Methods:
    * + diagnosePatient(): void
    * + prescribeMedication(): void

4. Appointment
  - Attributes:
    * - appointmentId: String
    * - appointmentDate: Date
    * - appointmentTime: String
    * - status: String
  - Methods:
    * + schedule(): void
    * + cancel(): void
    * + complete(): void

5. MedicalRecord

- Attributes:
  * - recordId: String
  * - diagnosis: String
  * - prescription: String
  * - dateCreated: Date
- Methods:
  * + addEntry(): void
  * + viewRecord(): void

6. Bill
- Attributes:
  * - billId: String
  * - totalAmount: Double
  * - paymentDate: Date
  * - paymentStatus: String
- Methods:
  * + generateInvoice(): void
  * + processPayment(): void

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3. RELATIONSHIPS, MULTIPLICITIES, AND EXPLANATIONS

1. Generalization / Inheritance:
- Patient inherits from Person
- Doctor inherits from Person
- Explanation: Person serves as the base class to model common demographic fields (name, phone, email) without code and data redundancy.

2. Association:
- Patient (1) ------- (0..*) Appointment
  * Explanation: One registered patient can schedule zero or many appointments over time, while each appointment is strictly booked for exactly one patient.
- Doctor (1) ------- (0..*) Appointment
  * Explanation: One doctor can conduct zero or many scheduled appointments, while each individual appointment is attended by exactly one doctor.

3. Composition:
- Patient (1) ◆------- (1..*) MedicalRecord
  * Explanation: Strong ownership relationship represented by the solid diamond on the Patient class. Medical records belong exclusively to one patient and cannot exist without that patient. If a patient record is deleted, their associated medical records are also removed.

4. Association:
   - Appointment (1) ------- (1) Bill
     * Explanation: Each completed consultation appointment generates exactly one bill for fee settlement.